Chung-Yen (Matt) Chiang

Software Engineer | Embedded Systems & Connectivity

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Professional Experience

Amazon | Connectivity Team Apr – Jun 2026
Software Development Engineer II

- Increased end-to-end Wi-Fi throughput to **2.36× the baseline** (10.59 to 25 Mbps) on FreeRTOS through memory and concurrency optimizations and CPU tuning.
- Designed runtime **NVRAM/CLM loading** from flash, enabling one firmware image to support multiple device variants and regulatory regions without per-region rebuilds.
- Led **Wi-Fi/BLE bring-up** on next-generation ASIC platforms; integrated wireless chipset BSPs and SDKs into production FreeRTOS firmware for new product introduction programs.

Amazon | Connectivity Team Aug 2023 – Mar 2026
Software Development Engineer I

- Designed and implemented a **modular BLE provisioning subsystem** with end-to-end encryption and dynamic country-code configuration, enabling global deployment from a shared firmware base.
- Optimized DHCP reconnection logic, reducing reconnection latency by **76%** under unstable network conditions.
- Tuned WMM/EDCA parameters to prioritize latency-sensitive traffic, reducing streaming p99 latency by **24%**.
- Implemented Wi-Fi power-saving and wake-up mechanisms, extending battery-operated IoT device standby from **40 to 65 days**.

Amazon | Connectivity Team Sep 2022 – Jul 2023
Software Development Engineer Intern

- Built automation for interoperability testing across **60+ commercial Wi-Fi router models**.

Institute of Information Science, Academia Sinica Jul – Aug 2022
Research Intern

- Implemented C and Python extensions to a malware-analysis sandbox to support detection of evasion techniques.

Technical Skills

Programming	C, C++, Python, Shell
Systems	Embedded firmware, FreeRTOS, BSP/SDK integration, platform bring-up
Networking	Wi-Fi, BLE, TCP/IP, DHCP, WMM/EDCA, IPsec
Tools	Linux, GDB / JTAG, perf, Wireshark, tcpdump, iperf3, Git
Languages	Mandarin (native), English (professional)

Education

Seoul National University 2026 – Present
M.S. in Computer Science and Engineering
Expected graduation: 2028

- Teaching Assistant, System Programming (Fall 2026): lead labs, hold TA office hours, answer student questions, and grade exams.

National Taiwan University of Science and Technology 2019 – 2023
B.B.A. in Information Management
GPA: 4.01 / 4.30

- Academic Excellence Award: top 2 of class for three consecutive semesters.
- MOST Undergraduate Research Scholarship (2021).

Selected Project

Secure Channels for Smart Manufacturing Jul 2021 – Feb 2022
Undergraduate Research Project, NTUST

- Implemented Site-to-Site IPsec connectivity between industrial gateways and AWS VPC; published a first-author paper at CISC 2022 (Taiwan).

[Paper, Chinese] [Repository]